Function & performance

National 5 · Design · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ describe how function influences the design of a product
- ☐ explain primary and secondary functions with examples
- ☐ describe how performance and fitness for purpose influence design
- ☐ describe maintenance issues associated with products
- ☐ explain how life expectancy affects design, manufacture and the environment

I DO — Worked example

Read how a full-mark answer is built

An office chair.

1 Describe how function has influenced the design of the office chair. (3 marks)

Points that earn marks from the marking instructions: one ✓ per point

- ✓ Seat height adjusts with a gas lift so it suits different desks
- ✓ Castors let the user move around the desk
- ✓ Swivel base lets the user turn without standing

Why marks are not awarded

- ✗ "The chair spins" / "it is adjustable" without saying how or why
- ✗ Lists of features

WE DO — Guided practice

Use the starters and word bank to help you

A wooden toy garage for young children.

2 Outline how maintenance has influenced the design of the toy garage. (1 mark)

Sentence starters

One … is …

Word bank

screws · replaced · tightened · surfaces · smooth · varnished · painted · standard

Tip: 1 marks = 1 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Making a phone battery hard to replace mainly affects which factor?
☐ Aesthetics ☐ Maintenance ☐ Target market ☐ Anthropometrics

b) A torch that also works as a bottle opener. The bottle opener is...
☐ Primary function ☐ Secondary function ☐ Performance ☐ Aesthetics

c) A razor with replaceable heads compared with a disposable razor has...
☐ Shorter life expectancy ☐ Longer life expectancy ☐ No maintenance ☐ No secondary function

4 Key terms: write the letter of the matching description next to each term.

1. **Primary function**	____	**A** How well a product meets its intended use, user, environment and life expectancy.
2. **Secondary function**	____	**B** Additional jobs that add value to the product.
3. **Fitness for purpose**	____	**C** The main job a product is designed to do.

5 Describe one maintenance consideration when designing a product. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?